

Installation of Open-Source EDA Tools

Introduction

This lab manual provides step-by-step instructions for installing and configuring various open-source EDA (Electronic Design Automation) tools required for RTL design, simulation, synthesis, physical design, verification, and layout analysis.

Objectives

After completing this lab, you will be able to:

1. Understand the setup process for open-source VLSI and ASIC design tools.
2. Install and configure commonly used EDA tools on Ubuntu/WSL2/Linux environments.
3. Verify successful installation of tools using terminal commands.
4. Understand dependencies required for OpenLane, OpenROAD, Verilator, Yosys, and related tools.
5. Prepare a complete open-source RTL-to-GDSII design environment.

Minimum System Requirements

Component	Minimum Requirement
Operating System	Ubuntu 22.04 LTS / Ubuntu 24.04 LTS / WSL2
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	Minimum 8 GB (16 GB Recommended)
Storage	At least 100 GB free disk space
Internet	Stable broadband connection
Virtualization	WSL2 or VirtualBox support (if using Windows)
Python	Python 3.10 or higher
Docker	Latest stable Docker version

Tools Covered in this Lab

The following open-source EDA tools are covered:

1. Yosys – RTL synthesis
2. Verilator – Fast Verilog/SystemVerilog simulation
3. Icarus Verilog for Verilog simulation
4. GTKWave – Waveform viewer
5. Magic – Layout editor
6. KLayout – GDS viewer and editor

7. OpenROAD – Physical design automation
8. OpenLane – RTL-to-GDSII flow
9. Netgen – LVS verification
10. Ngspice – Circuit simulation
11. Xschem – Schematic capture
12. OpenSTA – Static Timing Analysis

Installation Procedure

Step 1: Update System

Before starting, update your Ubuntu/WSL system:

```
sudo apt update && sudo apt upgrade -y
```

```
nielit@DESKTOP-NB1EVS0:~$ sudo apt update && sudo apt upgrade -y  
[sudo] password for nielit: |
```

Step 2: Install dos2unix utility and git

```
sudo apt-get install -y dos2unix git
```

```
nielit@DESKTOP-NB1EVS0:~$ sudo apt update && sudo apt upgrade -y  
[sudo] password for nielit: |
```

Step 3: Clone the Repository

Clone the repository to your system:

```
git clone https://github.com/mukulkuar011/EDA-Tools-Installer.git
```

```
nielit@DESKTOP-NB1EVS0:~$ git clone https://github.com/mukulkuar011/EDA-Tools-Installer.git
```

```
nielit@DESKTOP-NB1EVS0:~$ git clone https://github.com/mukulkuar011/EDA-Tools-Installer.git  
Cloning into 'EDA-Tools-Installer'...  
remote: Enumerating objects: 42, done.  
remote: Counting objects: 100% (42/42), done.  
remote: Compressing objects: 100% (40/40), done.  
remote: Total 42 (delta 12), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (42/42), 23.59 KiB | 805.00 KiB/s, done.  
Resolving deltas: 100% (12/12), done.  
nielit@DESKTOP-NB1EVS0:~$ |
```

Step 4: Move into Repository Directory

```
cd EDA-Tools-Installer
```

```
Resolving deltas: 100% (12/12), done.  
nielit@DESKTOP-NB1EVS0:~$ cd EDA-Tools-Installer/  
nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ |
```

Type ls command to see the contents of the directory:

ls

```

nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ ls
README.md  installfinalv2.sh  openlaneinstall.sh  pdk.sh
nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ |
    
```

Step 4: Make Scripts Executable

Give executable permissions to all shell scripts:

```
sudo chmod +x *.sh
```

```

nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ chmod +x *.sh
nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ |
    
```

Convert Windows line endings if required:

```
sudo dos2unix *.sh
```

```

nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ sudo dos2unix *.sh
dos2unix: converting file installfinalv2.sh to Unix format...
dos2unix: converting file openlaneinstall.sh to Unix format...
dos2unix: converting file pdk.sh to Unix format...
nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ |
    
```

Step 6: Run the Installer Script

Execute the installation script.

1. Install OpenLane Environment

```
./openlaneinstall.sh
```

```
nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ ./openlaneinstall.sh |
```

```

nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ ./openlaneinstall.sh
#=====#
# OpenLane Complete Installation Script                                     #
# Author      : Mukul Kumar                                              #
# Designation : Junior VLSI Engineer                                    #
# Organization : NIELIT CoE Noida                                         #
# Purpose     : Install Docker, OpenLane and Create Alias               #
# Compatible With:                                                         #
#   - Ubuntu Native                                                         #
#   - Ubuntu on VirtualBox                                                 #
#   - WSL2                                                                  #
#   - Multi-user Linux Labs                                                 #
#=====#
=====
Updating System Packages
=====
[sudo] password for nielit: |
    
```



```
Collecting mdurl==0.1 (from markdown-it-py>=2.2.0->rich<15,>=12->ciel<3,>=2.0.1->-r ./requirements.txt (line 3))
  Downloading mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Collecting typing_extensions>=4.5 (from anyio->httpx<0.29,>=0.22.0->ciel<3,>=2.0.1->-r ./requirements.txt (line 3))
  Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
  Downloading ciel-2.4.1-py3-none-any.whl (38 kB)
  Downloading click-8.4.0-py3-none-any.whl (116 kB)
  Downloading pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (807 kB)
    807.9/807.9 kB 24.5 MB/s 0:00:00
  Downloading httpx-0.28.1-py3-none-any.whl (73 kB)
  Downloading httpcore-1.0.9-py3-none-any.whl (78 kB)
  Downloading pcpp-1.30-py2.py3-none-any.whl (91 kB)
  Downloading rich-14.3.4-py3-none-any.whl (310 kB)
  Downloading pygments-2.20.0-py3-none-any.whl (1.2 MB)
    1.2/1.2 MB 26.0 MB/s 0:00:00
  Downloading zstandard-0.25.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (5.5 MB)
    5.5/5.5 MB 26.2 MB/s 0:00:00
  Downloading wheel-0.47.0-py3-none-any.whl (32 kB)
  Downloading h11-0.16.0-py3-none-any.whl (37 kB)
  Downloading markdown_it_py-4.2.0-py3-none-any.whl (91 kB)
  Downloading mdurl-0.1.2-py3-none-any.whl (10.0 kB)
  Downloading packaging-26.2-py3-none-any.whl (100 kB)
  Downloading anyio-4.13.0-py3-none-any.whl (114 kB)
  Downloading idna-3.15-py3-none-any.whl (72 kB)
  Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
  Downloading certifi-2026.4.22-py3-none-any.whl (135 kB)
Installing collected packages: pcpp, zstandard, typing_extensions, pyyaml, pygments, packaging, mdurl, idna, h11, click, certifi, whe
el, markdown-it-py, httpcore, anyio, rich, httpx, ciel
Successfully installed anyio-4.13.0 certifi-2026.4.22 ciel-2.4.1 click-8.4.0 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.15 markdow
n-it-py-4.2.0 mdurl-0.1.2 packaging-26.2 pcpp-1.30 pygments-2.20.0 pyyaml-6.0.3 rich-14.3.4 typing_extensions-4.15.0 wheel-0.47.0 zst
andard-0.25.0
PYTHONPATH= ./venv/bin/python3 -m pip freeze > venv/manifest.txt
./venv/bin/ciel enable --pdk sky130A
Version 0fe599b2afb6708d281543108caf8310912f54af not found locally, attempting to download...
  Downloading common.tar.zst... 100% 0:00:00
  Downloading sky130_fd_io.tar.zst... 100% 0:00:00
  Downloading sky130_fd_pr.tar.zst... 12% 0:00:01
```

```
[INFO]: Note that the following warnings have been generated:
[WARNING]: 1 warnings found by linter
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[WARNING]: Module sky130_fd_sc_hd__tapvprvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[WARNING]: There are max fanout violations in the design at the typical corner. Please refer to 'designs/spm/runs/openlane_test/repor
ts/signoff/31-rcx_sta.checks.rpt'.

Basic test passed
=====
Creating OpenLane Alias Command
=====
OpenLane command created successfully
=====

=====
OpenLane Installation Completed Successfully
=====

IMPORTANT:
Please logout and login again

Then launch OpenLane using:

openlane

===== SETUP COMPLETE =====
#####
# Author       : Mukul Kumar           #
# Designation  : Junior VLSI Engineer  #
# Organization : NIELIT CoE Noida      #
#####
nietlit@DESKTOP-NB1EV50:~/EDA-Tools-Installer$ |
```

This installs:

1. Docker
2. OpenLane
3. SKY130 PDK Support
4. OpenLane Launcher Command

2. Install Analog Design Environment

`./pdk.sh`

```
nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ ./pdk.sh |
```

```
nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ ./pdk.sh
#=====#
# Analog Complete with Open PDK Installation Script                                     #
# Author      : Mukul Kumar                                                            #
# Designation : Junior VLSI Engineer                                                    #
# Organization : NIELIT CoE Noida                                                       #
# Compatible With:                                                                     #
#   - Ubuntu Native                                                                    #
#   - Ubuntu on VirtualBox                                                             #
#   - WSL2                                                                              #
#   - Multi-user Linux Labs                                                            #
#=====#
=====
Updating System Packages
=====
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 https://download.docker.com/linux/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... 75%
```

```
Common install: Done.
Done.
=====
Creating xschem launcher wrapper...
=====
xschem launcher wrapper installed successfully.
=====
Creating Magic launcher wrapper...
=====
Magic launcher wrapper installed successfully.
===== INSTALLATION COMPLETE =====
===== FINAL TOOL STATUS =====
Magic
Location: /usr/local/bin/magic_bin
    ♦ Version: 8.3.644

Xschem
Location: /usr/local/bin/xschem_bin
    ♦ Version: XSCHEM V3.4.8RC

Netgen
Location: /usr/bin/netgen
    ♦ Version: NETGEN-6.2.2401
NETGEN-6.2.2401
NETGEN-6.2.2401

Ngspice
Location: /usr/local/bin/ngspice
    ♦ Version: *****

===== SETUP COMPLETE =====
#=====#
# Author      : Mukul Kumar                      #
# Designation : Junior VLSI Engineer              #
# Organization : NIELIT CoE Noida                  #
#=====#
nIELIT@DESKTOP-NB1EVSO:~/EDA-Tools-Installer$ |
```

This installs:

1. Ngspice
2. Xschem
3. Magic VLSI
4. Netgen
5. OpenPDKs (SKY130)

3. Install Digital RTL-to-GDSII Flow Tools

```
./installfinalv2.sh
```

```
nielit@DESKTOP-NB1EVSO:~/EDA-Tools-Installer$ ./installfinalv2.sh |
```

```
nielit@DESKTOP-NB1EVSO:~/EDA-Tools-Installer$ ./installfinalv2.sh
#=====#
# Author      : Mukul Kumar                      #
# Designation  : Junior VLSI Engineer             #
# Organization : NIELIT CoE Noida                 #
#=====#
Creating /labroot directory...
[sudo] password for nielit: |
```

```
[ 80%] Built target flags_program_name
[ 80%] Built target hashtablez_sampler
[ 80%] Building CXX object absl/flags/CMakeFiles/flags_config.dir/usage_config.cc.o
[ 81%] Building CXX object absl/container/CMakeFiles/raw_hash_set.dir/internal/raw_hash_set.cc.o
[ 82%] Linking CXX static library libabsl_cordz_handle.a
[ 82%] Built target cordz_handle
[ 83%] Linking CXX static library libabsl_vlog_config_internal.a
[ 84%] Linking CXX static library libabsl_cord_internal.a
[ 84%] Built target_vlog_config_internal
[ 84%] Building CXX object absl/log/CMakeFiles/log_globals.dir/globals.cc.o
[ 84%] Built target_cord_internal
[ 84%] Building CXX object absl/strings/CMakeFiles/cordz_info.dir/internal/cordz_info.cc.o
[ 85%] Linking CXX static library libabsl_flags_config.a
[ 85%] Built target_flags_config
[ 85%] Building CXX object absl/flags/CMakeFiles/flags_internal.dir/internal/flag.cc.o
[ 85%] Linking CXX static library libabsl_raw_hash_set.a
[ 85%] Built target_raw_hash_set
[ 85%] Linking CXX static library libabsl_cordz_info.a
[ 85%] Built target_cordz_info
[ 86%] Linking CXX static library libabsl_log_globals.a
[ 87%] Building CXX object absl/strings/CMakeFiles/cord.dir/cord.cc.o
[ 87%] Building CXX object absl/strings/CMakeFiles/cord.dir/cord_analysis.cc.o
[ 88%] Building CXX object absl/strings/CMakeFiles/cord.dir/cord_buffer.cc.o
[ 89%] Building CXX object absl/strings/CMakeFiles/cordz_sample_token.dir/internal/cordz_sample_token.cc.o
[ 89%] Built target_log_globals
[ 89%] Building CXX object absl/log/CMakeFiles/log_internal_log_sink_set.dir/internal/log_sink_set.cc.o
[ 89%] Building CXX object absl/log/CMakeFiles/log_initialize.dir/initialize.cc.o
[ 90%] Linking CXX static library libabsl_flags_internal.a
[ 90%] Built target_flags_internal
[ 90%] Linking CXX static library libabsl_log_initialize.a
[ 90%] Linking CXX static library libabsl_cordz_sample_token.a
[ 90%] Built target_log_initialize
[ 90%] Built target_cordz_sample_token
[ 90%] Linking CXX static library libabsl_log_internal_log_sink_set.a
[ 90%] Built target_log_internal_log_sink_set
[ 91%] Building CXX object absl/log/CMakeFiles/log_internal_message.dir/internal/log_message.cc.o
```




```
from /labroot/or-tools/ortools/flatzinc/fz.cc:39:
/labroot/or-tools/ortools/graph/iterators.h:106:19: warning: 'template<class _Category, class _Tp, class _Distance, class _Pointer, c
lass _Reference> struct std::iterator' is deprecated [-Wdeprecated-declarations]
106 | : public std::iterator<std::input_iterator_tag, IntegerType> {
    | ~~~~~^~~~~~
In file included from /usr/include/c++/13/bits/stl_iterator_base_funcs.h:66,
from /usr/include/c++/13/string:47,
from /usr/include/c++/13/bits/locale_classes.h:40,
from /usr/include/c++/13/bits/ios_base.h:41,
from /usr/include/c++/13/ios:44,
from /usr/include/c++/13/ostream:40,
from /usr/include/c++/13/iostream:41,
from /labroot/or-tools/ortools/flatzinc/fz.cc:24:
/usr/include/c++/13/bits/stl_iterator_base_types.h:127:34: note: declared here
127 | struct _GLIBCXX17_DEPRECATED iterator
    | ~~~~~^~~~~~
In file included from /labroot/or-tools/ortools/flatzinc/model.h:28,
from /labroot/or-tools/ortools/flatzinc/parser_main.cc:24:
/labroot/or-tools/ortools/graph/iterators.h:106:19: warning: 'template<class _Category, class _Tp, class _Distance, class _Pointer, c
lass _Reference> struct std::iterator' is deprecated [-Wdeprecated-declarations]
106 | : public std::iterator<std::input_iterator_tag, IntegerType> {
    | ~~~~~^~~~~~
In file included from /usr/include/c++/13/bits/stl_iterator_base_funcs.h:66,
from /usr/include/c++/13/string:47,
from /labroot/or-tools/ortools/flatzinc/parser_main.cc:18:
/usr/include/c++/13/bits/stl_iterator_base_types.h:127:34: note: declared here
127 | struct _GLIBCXX17_DEPRECATED iterator
    | ~~~~~^~~~~~
[100%] Linking CXX executable ../../bin/step_function_sample_sat
[100%] Linking CXX executable bin/fzn-cp-sat
[100%] Linking CXX executable bin/fzn-parser-cp-sat
[100%] Built target step_function_sample_sat
[100%] Linking CXX executable ../../bin/stop_after_n_solutions_sample_sat
[100%] Built target fzn
[100%] Built target fzn-parser_test
[100%] Built target stop_after_n_solutions_sample_sat
[sudo] password for nielit: |
```

```
199 |         columns = new Column*[ninputs];
    |         ^
/usr/include/c++/13/new: In function 'build_prefix_tree':
/usr/include/c++/13/new:128:26: note: in a call to allocation function 'operator new[]' declared here
128 | _GLIBCXX_NODISCARD void* operator new[](std::size_t) _GLIBCXX_THROW (std::bad_alloc)
    | ~~~~~^~~~~~
/labroot/OpenROAD/third-party/abc/src/aig/gia/giaEquiv.c: In function 'Gia_ManTransferEquivs2':
/labroot/OpenROAD/third-party/abc/src/aig/gia/giaEquiv.c:2794:20: warning: argument 1 range [18446744071562067968, 184467440737095516
15] exceeds maximum object size 9223372036854775807 [-Walloc-size-larger-than=]
2794 |     pOld->pReprs = ABC_CALLOC( Gia_Rpr_t, Gia_ManObjNum(pOld) );
    |                        ^
/usr/include/stdlib.h:675: note: in a call to allocation function 'calloc' declared here
675 | extern void *calloc (size_t __nmemb, size_t __size)
    | ~~~~~^~~~~~
In function 'Vec_PtrAllocExact',
inlined from 'Bac_NtkTransformToPtrBoxes' at /labroot/OpenROAD/third-party/abc/src/base/bac/bacPtrAbc.c:423:43:
/labroot/OpenROAD/third-party/abc/src/misc/vec/vecPtr.h:103: warning: argument 1 range [18446744056529682432, 18446744073709551608] e
xceeds maximum object size 9223372036854775807 [-Walloc-size-larger-than=]
103 |     p->pArray = p->nCap? ABC_ALLOC( void *, p->nCap ) : NULL;
    | ~~~~~^~~~~~
/usr/include/stdlib.h: In function 'Bac_NtkTransformToPtrBoxes':
/usr/include/stdlib.h:672:14: note: in a call to allocation function 'malloc' declared here
672 | extern void *malloc (size_t __size) __THROW __attribute_malloc__
    | ~~~~~^~~~~~
In function 'Vec_PtrAllocExact',
inlined from 'Bac_NtkTransformToPtrBoxes' at /labroot/OpenROAD/third-party/abc/src/base/bac/bacPtrAbc.c:423:43,
inlined from 'Bac_NtkTransformToPtr' at /labroot/OpenROAD/third-party/abc/src/base/bac/bacPtrAbc.c:455:50:
/labroot/OpenROAD/third-party/abc/src/misc/vec/vecPtr.h:103: warning: argument 1 range [18446744056529682432, 18446744073709551608] e
xceeds maximum object size 9223372036854775807 [-Walloc-size-larger-than=]
103 |     p->pArray = p->nCap? ABC_ALLOC( void *, p->nCap ) : NULL;
    | ~~~~~^~~~~~
/usr/include/stdlib.h: In function 'Bac_NtkTransformToPtr':
/usr/include/stdlib.h:672:14: note: in a call to allocation function 'malloc' declared here
672 | extern void *malloc (size_t __size) __THROW __attribute_malloc__
    | ~~~~~^~~~~~
[100%] Built target openroad
[sudo] password for nielit: |
```



```
Verilator
Location: /usr/local/bin/verilator
  * Version: Verilator 5.038 2025-07-08 rev v5.038

GTKWave
Location: /usr/bin/gtkwave
  * Version: GTKWave Analyzer v3.3.116 (w)1999-2023 BSI

Yosys
Location: /usr/bin/yosys
  * Version: Yosys 0.33 (git sha1 2584903a060)

OpenROAD
Location: /usr/local/bin/openroad
  * Version: OpenROAD 26Q2-1311-gd4c534c106
26Q2-1311-gd4c534c106

Klayout
Location: /usr/bin/klayout
  * Version: KLayout 0.28.16

OpenSTA
Location: /usr/local/bin/sta
  * Version: OpenSTA 3.1.0 76c4d6df35 Copyright (c) 2026, Parallax Software, Inc.
3.1.0

RISC-V Tool Chain
Location: /usr/bin/riscv64-unknown-elf-gcc
  * Version: riscv64-unknown-elf-gcc (13.2.0-11ubuntu1+12) 13.2.0

===== SETUP COMPLETE =====
#####
# Author       : Mukul Kumar           #
# Designation  : Junior VLSI Engineer  #
# Organization : NIELIT CoE Noida      #
#####
nielit@DESKTOP-NB1EVS0:~/EDA-Tools-Installer$ |
```

This installs:

1. Yosys
2. Verilator
3. GTKWave
4. OpenROAD
5. OpenSTA
6. KLayout
7. RISC-V Toolchain
8. Icarus Verilog

Notes and Precautions

1. Use Ubuntu-based systems for best compatibility.
2. Installation may take significant time depending on internet speed.
3. Some tools require Docker and virtualization support.
4. Ensure at least 100 GB free storage before installation.
5. Avoid interrupting the installation process midway.

Conclusion

In this lab manual, the installation procedure for multiple open-source EDA tools was successfully explained using the EDA-Tools-Installer repository. The environment prepared through this setup can be used for RTL design, simulation, synthesis, verification, physical design, and layout analysis using fully open-source workflows.